Electrical And Electronics Engineering

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Unit wise important big questions for university examination.
Unit 1
1. Numerical problem on Mesh analysis.
2. Numerical problem on Nodal analysis.
3. Numerical problem on Thevenin’s theorem.
4. Numerical problem on Superposition theorem.
5. Numerical problem on Maximum power transfer theorem.
6. Numerical problem on R-L,R-C and R-L-C circuits. (Finding resistance (R), inductance (L)inductive reactance (XL), capacitive reactance (XC), impedance (Z) , power factor (pf), real power (P), reactive power (Q),
7. Deriving the equation for RMS value, average values of AC waveforms (sine wave, half rectified sine wave, full rectified sine wave)
Unit 2
1. Describe the construction and operation of JFET. Also draw its characteristics.
2. Describe the construction and operation of MOSFET. Also draw its characteristics
3. Describe the construction and operation of BJT. Also draw its characteristics
4. Describe the construction and operation of SCR. Also draw its characteristics.
5. Simplification of Boolean expression using Boolean Laws and theorems.
6. Simplification of Boolean expression using K map.
Unit 3
1. Describe the construction and working of DC generator with neat diagram.
2. Describe the construction and working of DC motor with neat diagram.
3. Describe the construction and operation of single-phase transformer with neat diagram.
4. Describe the construction and working of three phase induction motor with neat

diagram.
5. Describe the construction and working of BLDC motor with neat diagram.
6. Describe the construction and working of stepper motor with neat diagram.
7. Describe the construction and working of servo motor with neat diagram.

Unit 4

1. Describe the construction and working of Permanent magnet moving coil (PMMC) instrument with a neat diagram. Also give the torque equation.
2. Describe the construction and working of Moving iron (MI) instrument with a neat diagram. (Both attraction type and repulsion type)
3. Describe the construction and working of Dynamometer type of instrument with a neat diagram.
4. Describe the construction and working of Linear variable differential transformer (LVDT)/ Inductive transducer / with neat diagram. Also draw the graph between displacement and output voltage.
5. Describe he construction and working of capacitive transducer with a neat diagram.
6. Describe the block diagram and working of liquid crystal display (LCD).

Unit 5

1. Draw and explain the layout of generation, transmission and distribution of power.
2. List and explain all the substation equipment.

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3. Draw and explain the key diagram of 11kV/440 V indoor substation with neat diagram.
4. Explain different types of earthing schemes and list the importance of earthing.
5. Explain different types of Electric vehicles with its advantages and disadvantages.

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[Page 1]
Reg. No.
I
Time: 3 hours Max. Marks: 75
Marks BL CO PO
How many loops are there in the given circuit?1. i 3 1 2
wa 15Q
A B
6Q an
1 2 1 12.
Series circuit: Voltage division rule
Parallel circuit: Current division rule
Open circuit: Resistance is zero
Short circuit: Resistance is maximum
How many of the above pairs are correctly matched?
3 1i iConsider the following Statements3.
How many statements given above are correct?
19DF1-21EES101T
A)3
C)1
B)5

D)2
B)4
D)2
B)3
D)4
1. In a parallel circuit, the current is the same through all the elements
2. Current will not flow in an open circuit
3. Ohm’s law does not apply to parallel circuits
4. In a series circuit, voltage is the same across all the resistors
B.TECH/M.TECH(INTEGRATED) DEGREE EXAMINATION, DECEMBER 2024
First Semester
21EES101T - ELECTRICAL AND ELECTRONICS ENGINEERING
(For the candidates admitted from the academic year 2024-2025)
PART - A (20 x 1 = 20 Marks)
Answer ALL Questions
A/x/sz-
5Q
A)3
C)4
Consider the following pairs
A) 2
C)1

+
lOOVf
Note:
(i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall
invigilator at the end of 40th minute.
(ii) Part - B & Part - C should be answered in answer booklet.

1 11 1
4.
22 21
5.
ii 21to control6.
22 11
7.
2 11 1
8.
3 11 1
9.
ii 2 3

10.
1 2 3 1
11.
1 2 3 112.
2 4 113.
i 4 1114.
2 4 115.
2 4 1116.

impuritie is tri-valent.
A) Antimony
C) Boron
A) Chopper
C) Servomotor
ensures
deflected position
A) Deflecting torque
C) Inertia torque
B)fuse
D) instrument
B) Base , Emitter, Collector
D) Anode, Gate, Cathode
B) Photoemissive cell
D) Thermistor
SCR terminals are
A) Anode, Gate, Collector
C) Gate , Emitter, Collector
B) Reverse biased condition
D) Cutoff
A)Sensor
C)UPS
is an electronically modulated optical device
A)Proximity Sensor B)LCD
C) Optocoupler D) IR sensor
The expression for RMS voltage in half wave rectifier is
A)2Vm/7T B)Vm/2
C)Vm/7t D)Vm/27r
What does the equivalent circuit of the diode shown represent?
i—01—
A) Forward biased condition
C) Open circuit
MOSFET is a type of transistor that uses an
A) amplifier
C) electric field
A) Brushes

C) Split rings
is a polyphase synchronous motor with a pennanent magnet rotor.
A)Brushless DC motor B) Induction motor
C) Transformer D) DC motor
is a combination of an electric motor, a transmission, and a control system
B) Transducer
D) Electric drive
is a motor that can rotate with great precision
B) Crane
D) DC motor
that the moving system takes just the required time to reach its final 1
B) Damping torque
D) Spring torque
is a device which converts the energy from one form to another form
A)DSO . B) Multimeter
C) Transducer D) Thermocouple
Which device works on the simple principle of change in resistance due to a change in 1
temperature?
A)Photoconductive cell
C)LVDT
B) Arsenic
D) Phosphorous
of the DC generator is in the form of a laminated slotted drum.
B) Armature
D)Pole shoe
THE HELPERS

[Page 3]
2 11 517.
2 5 1118.
i 2 5 119.
i 2 5 120.
Marks BL CO PO
28 3 121 a. Find the current through 2 ohms in the given circuit using superposition
theorem.
20 Q
■')4A
8 2 1 2
8 2 2 1
8 2 2 1
8 2 3 1
8 2 3 1
8 2 4 1
8 3 4 1
8 2 5 1

8 2 5 1

conductors
A) Cross arms
C) Insulators
fault conditions.
A) Isolator
C) Busbar
10 Q
-'VAV'-
20
AV/A
B) Circuit breaker
D) Power transformer
B)Bulk
D) Offshore
is an equipment which can open or close a circuit under normal as well as
PART - B (5 x 8 = 40 Marks)
Answer ALL the Questions
(OR)
b. Derive the RMS and average voltage expression for a sinusoidal waveform.
22 a. Explain the construction ,working and characteristics of SCR with neat diagram
(OR)
b. Discuss the operation of switched mode power supply with neat diagram
23 a. Describe the construction and working of a three-phase induction motor with a neat diagram.
(OR)
b. Discuss the operation of chopper-fed DC drive with a neat diagram.
24 a. Explain the construction and operation of a permanent magnet moving coil instrument with a neat diagram
(OR)
b. Discuss the different types of sensors and their application in smart buildings.
25 a. Write short notes on battery electric vehicle and hybrid electric vehicle with diagram.
(OR)
b. With a neat diagram, describe the 11 kV/400 V indoor substation.
5Q
)40V
4 I
20 vO
$30
A large number of solar cells are interconnected to form
A)battery B) photovoltaic cell
C) solar inverter D) solar module
are attached to the poles or towers and are used to support the

B) Conductors
D)Fuse
transmission system is more economical when the distance of
transmission is more than 600 km.
A) AC
C)HVDC
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[Page 4]
Marks BL CO PO
15 3 1 2
£8 £24V~4r_
2 215 3Simplify the given boolean expressions using K-map and implement using logic gates.27
******
Page 4 of 4 19DF1-21EES101T
26 Determine the current through all the resistors using nodal voltage analysis in the given circuit.
(i) Y(A,B,C,D)= Em(0,2,5,7,8,9,10,11,12,13,14,15)
(ii) Y (A, B, C, D) = Km (1,2,5,6,7,9,12,13)
5£2
—■AW-
12 £2
WvV----
10 <2 B
MV—r
PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Questions
U- GV
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PN Junction

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[Page 1]
SRMINSTITUTEOFSCIENCEANDTECHNOLOGY-RAMAPURAMDEPARTMENTOFECE18EES101J–BASICELECTRICALANDELECTRONICSENGINEERINGMULTIPLECHOICEQUESTION-QUESTIONBANK(NOTE:Thisquestionbankhas180questionsOutofwhich18easyquestions(bloomslevel:Remembering,understanding)ineachunit18*5=9012moderatequestions(bloomslevel:Applying,Analyzing)ineachunit12*5=606toughquestions(bloomslevel:Evaluating,Creating)ineachunit6*5=30)
UNIT1-ELECTRICALCIRCUITS
EASYQUESTIONS
1)Theveninresistanceisfoundby________A.ShortingallvoltagesourcesB.OpeningallcurrentsourcesC.ShortingallvoltagesourcesandopeningallcurrentsourcesD.OpeningallvoltagesourcesandshortingallcurrentsourcesANSWER:C
2)Inastarconnectedsystem,thecurrentflowingthroughthelineisA.GreaterthanthephasecurrentB.EqualtothephasecurrentC.LesserthanthephasecurrentD.zero
ANSWER:B
3)The2ohmand3ohmresistorareinseriestheequivalentresistanceisA.1.2B.5C.4.2D.1.4ANSWER:B
4)TheinternalresistanceforthemaximumtransferofpowershouldbeA.equaltoloadresistanceB.greatert

hanloadresistanceC.zeroD.lesserthanloadresistanceANSWER:A
5)IfthevoltagefrequencyappliedtoaseriesRCcircuitisincreased,thenthephaseanglewill

[Page 2]
A.IncreasesB.reducesC.remainsthesameD.zeroANSWER:A
6)InanRLCcircuitabovetheresonantfrequency,
thecurrentwillA.lagstheappliedvoltageB.leadstheappliedvoltageC.isinphasewiththeappliedvoltagesD.iszeroANSWER:A
7)TheequationforohmslawisA.V=IR,atcontanttemperatureB.V=ICC.V=ILD.V=I/RANSWER:A
8)A6kHzsinusoidalvoltageisappliedtoaseries RC
circuit.ThefrequencyofthevoltageacrosstheresistorisA.6KhzB.12KhzC.13KhzD.14KhzANSWER:A
9)Inacertainload,theactualpoweris150Wandthereactivepoweris125VAR.Whatistheapparentpower?
A.19.52WB.195.2WC.375WD.24WANSWER:B
10)Whatistheunitofpower?A.WattB.NewtonC.JouleD.HenryANSWER:A

[Page 3]
11)MeshanalysisemploysthemethodofA.KVLB.KCLC.BothKVLandKCLD.NeitherKVLorKCLANSWER:A
12)Ifthereare10nodesinacircuit,howmanyequationsdoweget?A.10B.9C.8D.7ANSWER:B
13)SuperpositiontheoremcanonlybeusedforcircuitsA.ElementresistiveB.ElementpassiveC.LinearbilateralelementsD.Non-linearelementsANSWER:C
14)Eachphaseofathreephasealternatordeltaconnectedproducesavoltageof11KVandacurrentof1000Aatpf0.9.Findlinevoltageandlinecurrent.A.11KV,1732AB.11KV,1632AC.3.33KV,1732AD.3.33V,1000AANSWER:A
15)Inabalancedthreephasesystemthreevoltagesdifferin____electricalfromeachotherinasequenceandhaveequalmagnitude.A.240B.120C.360D.0ANSWER:A
16)Forseriescircuittheequivalentresistanceis___thegreatestresistanceconnectedinseriescircuit.A.lesserthanB.greaterthanC.equaltoD.notequaltoANSWER:A

[Page 4]
17)Thenon-linearcircuitparametersare?
A.InductanceB.CapacitanceC.ResistanceD.TransistorANSWER:A
18)Inaseries RC circuit,findtheRMSvoltagewherethevoltageacrossresistoris12V(rms)
andvoltageacrosscapacitoris15V(rms)
.ThermssourcevoltageisA.3B.27C.19.2D.40ANSWER:C
MODERATEQUESTIONS
1)ThevoltageVusingnodalanalysis
A.-60VB.60VC.-40VD.40VANSWER:A
2)FindthecurrentflowingbetweenterminalsAandBofthecircuitshownbelow.
A.1B.2C.3

[Page 5]
D.4ANSWER:D
3)FindthecurrentflowingbetweenterminalsAandB.
A.1B.2C.3D.4ANSWER:D

4)CalculatethetotalresistancebetweenthepointsAandB.
A.7ohmB.4ohmC.7.6ohmD.0.48ohmANSWER:C
5)CalculatetheequivalentresistancebetweenAandB.

[Page 6]
A.2B.4C.6D.8ANSWER:B
6)Theresistanceareconnectedinseries.Findtheequivalentresistance
A.35B.25C.15D.5ANSWER:D
7)Anelectrickettlehasaresistanceof30ohm.Whatcurrentwillflowwhenitisconnectedto240Vsupply.Alsofindthepower.A.8A,1.92KwB.9A,3KwC.10A,4KwD.12A,5KwANSWER:A
8)AnidealvoltagesourcehasA.ZerointernalresistanceB.OpencircuitvoltageequaltothevoltageonfullloadC.TerminalvoltageinproportiontocurrentD.TerminalvoltageinproportiontoloadANSWER:A
9)Tofindimpedanceintheveninstheorem.A.AllindependentcurrentsourcesareshortcircuitedandindependentvoltagesourcesareopencircuitedB.AllindependentvoltagesourcesareopencircuitedandallindependentcurrentsourcesareshortcircuitedC.Allindependentvoltageandcurrentsourcesareshortcircuited

[Page 7]
D.AllindependentvoltagesourcesareshortcircuitedandallindependentcurrentsourcesareopencircuitedANSWER:A
10)ApplicationofNorton’stheoremtoacircuityieldsA.EquivalentcurrentsourceandimpedanceinseriesB.EquivalentcurrentsourceandimpedanceinparallelC.EquivalentimpedanceD.EquivalentcurrentsourceANSWER:A
11)Whatwillbetheresistanceofthewirewhichhas0.14mmdiameterandspecificresistance9.6microohm-cmis440cmlong.TheresistanceofthewirewillbeA.9.6ohmB.11.3ohmC.13.7ohmD.27.4ohmANSWER:D
12)InSuperpositiontheorem,whileconsideringasource,allothervoltagesourcesare?
A.opencircuitedB.shortcircuitedC.changeitspositionD.removedfromthecircuitANSWER:B
TOUGHQUESTIONS
1)FindthevalueofthecurrentsI1,I2andI3flowingclockwiseinthefirst,secondandthirdmeshrespectively.
A.1.54A,-0.189A,-1.195AB.2.34A,-3.53A,-2.23AC.4.33A,0.55A,6.02AD.-1.18A,-1.17A,-1.16AANSWER:A

[Page 8]
2)CalculatethemeshcurrentsI1andI2flowinginthefirstandsecondmeshesrespectively
A.1.75A,1.25AB.0.5A,2.5AC.2.3A,0.3AD.3.2A,6.5AANSWER:A
3)FindthenodevoltageV.
A.1VB.2VC.3VD.4VANSWER:D
4)FindthevalueofVifthecurrentinthe3ohmresistor=0.
A.3.5VB.6.5VC.7.5VD.8.5VANSWER:B
5)Inthecircuitshown,findthecurrentthrough4ΩresistorusingSuperpositiontheorem.

[Page 9]
A.4B.5C.6D.7ANSWER:B
6)FindthevalueofV1andV2.
A.87.23V,29.23VB.23.32V,46.45VC.64.28V,16.42VD.56.32V,78,87VANSWER:C
UNIT2-D.CMACHINES&A.CMACHINES

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EASYQUESTIONS
1.PowerfactorofZeroindicatesa)Purelyresistiveelementc)combinationofboth(A)and(B)
b) Purelyinductiveelementd)PurelycapacitiveelementAnswer:B
2.Noloadspeedofwhichofthefollowingmotorwillbehighesta)Shuntmotorc)Differentiallycompoundmotor
b)Cumulativelycompoundmotord)SeriesmotorAnswer:D
3.Whichofthefollowingisthemosteconomicalmethodofstartingasinglephasemotor?
a)Resistancestartmethodc)Capacitancestartmethod
b)Inductancestartmethodd)SplitphasemethodAnswer:C
4.Materialusedfortheconstructionoftransformercoreisusuallya)wood b)copper c)Aluminium d)SiliconsteelAnswer:D
5.Thepowerfactorinresistivecircuitisa)0.6pflagging b)0.8pflagging c)0.8pflagging d)1Answer:D
6.DCgeneratorworksontheprincipleofa)Fleming'srighthandrulec)Faraday'slaw
b)Fleming'slefthandruled)Lenz'slawAnswer:A
7.Twowindingofatransformerare------coupledA)Magneticallyc)Bothelectricallyandmagnetically
b)Electricallyd)ResistivelyAnswer:A
8.Thesynchronousspeedofa4poleinductionmotorfor50hzpowersupplyis------rpm.a)1500 b)1000 c)750 d)1440Answer:A
9.Powerfactoristheratioofa)Impedancetoresistancec)Resistancetoimpedance
b)Resistancetoreactanced)ReactancetoimpedanceAnswer:C

[Page 11]
10.Formfactoristheratioofa)MaximumtoRMSValuec)RMStoaveragevalue
b)Maximumtoaveragevalued)RMStomaximumvalueAnswer:C
11.Theunitofmagneticfluxdensityisa)Henry/meter b)Tesla c)Amp/meter d)volt/meterAnswer:B
12.ThefluxisanalogoustoA)Voltageinelectriccircuitc)Powerinelectriccircuit
b)Currentinelectriccurrentd)ResistanceinelectriccurrentAnswer:B
13.Whichmotorisconstantspeedmotor?a)DCseriesmotor b)DCshuntmotor c)Dccompoundmotor d)InductionmotorAnswer:B
14.Theprimarywindingofatransformerhas110vacrossit.Whatisthesecondaryvoltageiftheturnsratiois8?a)8.8V b)88V c)880V d)8800VAnswer:C
15.Amagnetizingforceof8000A/misappliedtoacircularmagneticcircuitofmeandiameter30cmbypassingacurrentthroughacoilwoundonthecircuitis750turned.Ifthecoilisuniformlywound,calculatethecurrentflowinthecircuit.a)10.05A b)9.8A c)11A d)12AAnswer:A
16.Whatwillbethemagneticpotentialdifferenceacrosstheairgapof2cmlengthinmagneticfieldof200AT/m?a)2AT b)4AT c)6AT d)10ATAnswer:B

17.Asingle-windingsingle-phasemotorhasa)Lowstartingtorquec)Highstartingtorque
b)zerostartingtorqued)Startingtorqueequaltofull-loadtorque.Answer:B
18.Adifferentiallycompoundedmotorunderhigh-over-loadconditionsbehavelikeaana)Shuntmotor
b)Seriesmotor c)Cumulativecompoundmotor d)SynchronousmotorAnswer:B

MODERATEQUESTIONS1.
Anelectricmotorwithconstantpowerwillhaveatorquespeedcharacteristicintheformofaa)Straightline throughtheoriginc)Circleabouttheorigin
b)Straightlineparalleltothespeedaxisd)RectangularhyperbolaAnswer:B
2.IfloadcurrentandfluxofDCmotorareheldconstantandvoltageappliedacrossit'sarmatureisincreaseby5%,thenspeedofmotorwilla)Increasebyabout5% b)Reducebyabout5% c)Remainunaltered
d)DependsonotherfactorsAnswer:A
3.Theslipofaninductionmotornormallydoesnotdependona)Rotorspeed b)Synchronousspeed
c)Shafttorque d)Core-losscomponentAnswer:D
4.A4-
Pointstarterisusedtostartandcontrolthespeedofaa)DCshuntmotorwitharmatureresistancecontrolc)DCseriesmotor
b)DCshuntmotorwithfieldweakeningcontrold)DCcompoundmotorAnswer:A
5.Thedcmotor,whichcanprovidezerospeedregulationatfullloadwithoutanycontroller,isa)Series
b)Shunt c)Cumulativecompoundd)DifferentialcompoundAnswer:B
6.ASolenoidiswoundwithacoilof100turns.Thecoilisoflength50cmandiscarryingacurrentof2A.Determinethemagneticfieldstrengthatthelineofthesolenoid.a)450AT/m b)400AT/m c)500AT/m
d)600AT/mAnswer:B
7..Ifthecross-
sectionalareaofamagneticfieldincreases,butthefluxremainsthesame,thefluxdensitya)Increases
b)Decreases c)Remainsthesame d)DoublesAnswer:B
8.Whatisthereluctanceofamaterialthathasalengthof0.07m,across-
sectionalareaof0.014m2,andapermeabilityof4,500Wb/At×m?a)1111At/Wb b)111At/Wb c)11At/Wb
d)1At/WbAnswer:A
9.A47Ohmresistorandacapacitorwithacapacitivereactanceof120
areinseriesacrossanacsource.Whatisthecircuitimpedance,Z

a)126ohm b)127ohm c)128ohm d)129ohmAnswer:D
10.Theabilityofamaterialtoremainmagnetizedafterremovalofthemagnetizingforceisknownasa)Permeability b)Reluctance c)Retentivity d)HysteresisAnswer:C
11.Theinducedvoltageacrossastationaryconductorinastationarymagneticfieldisa)Zero b)Increased
c)Decreased d)ReversedinpolarityAnswer:A
12.ADCgeneratorisrotatedat50revolutions/sec.howmanytimesdoesthedcoutputvoltagereachmaximumineachsecond?a)50 b)100 c)150 d)3000Answer:B
TOUGHQUESTIONS
1.Ina
seriesRCcircuit,12Vismeasuredacrosstheresistorand15Vismeasuredacrossthecapacitor.Thesourcevoltageisa)3V b)27V c)19.2V d)12VAnswer:C

2.Eachphaseofa3phasestarconnectedalternatorproducesavoltageof11000Vandcurrentof1000Aatpowerfactor0.9.findlinevoltage,linecurrentandtotalcapacityofthealternator.a)VL=19053V,IL=1000A,Capacity=29.7MWc)VL=19053V,IL=1000A,Capacity=29.7MW
b)VL=2000V, IL=1500A,Capacity=25MWd)VL=2500V,IL=500A,Capacity=35MWAnswer:A
3.ADCmotortakesanarmaturecurrentof110Aat480V.Thearmaturecircuitresistanceis.2ohm.themachinehas6polesandthearmatureislapconnectedwith864conductors.thefluxperpoleis0.05wb.calculatespeedandtorquedevelopedbythearmature.a)N=630rpm&T=750N-mc)N=636rpm&T=756N-m
b)N=635rpm&T=786N-md)N=536rpm&T=856N-mAnswer:C
4.Theregulationofdcgeneratoronfullloadisabouta)15to20% b)20to25% c)10to15% d)5to10%Answer:D
5.Forasingle-phasecapacitorstartinductionmotorwhichofthefollowingstatementsisvalid?

[Page 14]
a)Thecapacitorisusedforpowerfactorimprovementc)Thedirectionofrotationcannotbechanged
b)Thedirectionofrotationcanbechangedbyreversingthemainwindingterminalsd)Thedirectionofrotationcanbechangedbyinterchangingthesupplyterminals Answer:B
6.ADCseriesmotorhaslinearmagnetizationandnegligiblearmatureresistance,themotorspeedisa)Directlyproportionalto√T
b)Inverselyproportionalto√T
c)DirectlyproportionaltoT
d)Inverselyproportionalto√TAnswer:B

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UNIT3-ELECTRONICDEVICES
EASYQUESTIONS
1.Thefunctionofchokeandstarterinafluorescentlampcircuitistoa)Reducethepowerconsumedbythefluorescentlampb)Createahighvoltageacrossthetubeduringstartingc)Helptodrawveryhighcurrentduringstartingd)ImprovethepowerfactorofthefluorescentlampcircuitAnswer:B
2.Movingcoilinstrumentscanbeusedona)DConly b)sinusoidalAConly c)AllACwaveforms
d)ACandDcbothAnswer:A
3.PMMCInstrumentareusedfor-----quantitymeasurementa)AC b)Magnetic c)DC d)BothACandDCAnswer:C
4.MovingPartsofinstrumentsaresupportedina)Bushbearings b)Ballbearings c)RollerBearings
d)JewelledbearingsAnswer:D
5.Asinglelampcontrolledbytwo-wayswitchesattwoplacesiscalleda)Staircasewiring
b)Corridorwiring c)Cleatwiring d)BatterwiringAnswer:A
6.Inamovingcoilammeter,
thedeflectingtorqueisdirectlyproportionaltothea)Squareofthecurrenttobemeasuredb)Currenttobemeasuredc)Twicethecurrenttothemeasuredd)SquarerootofthecurrenttobemeasuredAnswer:B
7.Whichcannotreducetheearthresistance?
a)Pouringwaterintheearthpitc)Increasingthedepthoftheearthpit
b)Decreasingplatearead)connectingelectrodesinparallel

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Answer:B
8.Theearthplatemadeupofa)copper b)aluminium c)lead d)ironAnswer:A
9.Goodearthingisthatwhichgivesa)verylowresistance b)Highresistance c)Equalresistance

d)zeroresistanceAnswer:A
10.Thehightorquetoweightratioinananalogindicatinginstrumentindicatesa)Highfrictionlossc)Nothingasregardsfrictionloss
b)Lowfrictionlossd)CopperlossAnswer:B
11.GaAs,LEDemitsradiationinthe(a)UVregion(b)Bluecolor(c)visibleregion (d)infra-redregionAnswer:D
12.Theripplefactorofbridgerectifieris(a)0.482 (b)0.812(c)1.11(d)1.21Answer:A
13.Thebasicpurposeoffilteristo(a)minimizevariationsina.c.inputsignal(b)suppressharmonicsinrectifiedoutput(c)removesripplesfromrectifieroutput (d)stabilizedcoutputvoltageAnswer:C
14.IfVmisthepeakvalueofanappliedvoltageinhalfwaverectifierwithalargecapacitoracrossload,thenPIVis(a)Vm/2 (b)Vm (c)2Vm(d)1.414VmAnswer:B
15.JunctionbreakdownofaPNjunctionoccurs(a)withforwardbias
(b)withreversebias(c)becauseofmanufacturingdefect(d)NoneofaboveAnswer:B
16.InPNjunctiondiodedynamicconductanceisdirectlyproportionalto(a)theappliedvoltage(b)temperature(c)thecurrent (d)thethermalvoltageAnswer:C

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17.Inafullwaverectifier,
thecurrentineachofthediodesflowsfor(a)completecycleoftheinputsignal(b)halfcycleoftheinputsignal(c)lessthanhalfoftheinputsignal(d)NoneofaboveAnswer:A
18.WhenthePNjunctiondiodeisforwardbiased(a)theonlycurrentisholecurrent(b)theonlycurrentiselectroncurrent(c)theonlycurrentisproducedbymajoritycarriers(d)thecurrentisproducedbybothholesandelectronsAnswer:C
MODERATEQUESTIONS
1.For1N4736ZenerdiodehasZz=3.5Ω.ThedatasheetgivesVzt=6.8VatIzt=37mA,Whatisvoltageacrosszenerterminalswhenthecurrentis50mA?(a)6.85V (b)7.85V(c)8.85V(d)9.95VAnswer:A
2.ASiPNjunctionhasareversesaturationcurrentofI0=30nAatroomtemperature,thejunctionforwardvoltagerequiredtoproducecurrentof0.1mAis(a)0.42V (b)0.55V(c)0.80V(d)0.49VAnswer:A
3.Thevalueofreversebiasresistanceforanidealdiodeis__________(a)infinity
(b)o(c)one(d)noneoftheaboveAnswer:A
4.Semiconductormaterialhave__________temp.coefficient(a)Positive (b)Negative
(c)Bothpositiveandnegative (d)NoneAnswer:B
5.Azenerdiodeworksontheprincipalof(a)tunnelingofchargecarriersacrossjunction(b)thermionicemission(c)diffusionofchargecarriersacrossjunction(d)hoppingofchargecarriersacrossjunctionAnswer:C
6.Whichoneofthefollowingtypesofindicatinginstrumentisanelectometer?
a)Electrodynamometerb)PMMC c)Electrostatic d)MovingironAnswer:C

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7.Incleatwiringtheporcelainareveryeasytoerectandfixedatadistanceofa)4.5cmto15cmapart
b)5.5cmto20cmapart c)6.5cmto25cmapart d)7.5cmto30cmapartAnswer:A
8.Influorescentlampthelightoutputis----lumensperwatt.a)70 b)80 c)90 d)95Answer:A
9.Thedeviceusedinserieswiththelinewireisa)C.B b)isolator c)Fuse d)BothC.BandisolatorAnswer:C
10.Theearth'spotentialisalwaysa)Zero b)one c)Lesserthanone d)GreaterthanoneAnswer:A
11.Iftheinputsupplyfrequencyis50Hz,theoutputsupplyfrequencyofabridgewaverectifieris_________

__Hz(a)100 (b)75(c)50(d)25Answer:A

12.Ahalfwaverectifierhasaninputvoltageof240Vrmsifthestepdowntransformerhasturnsratioof8:1,whatistheloadvoltage?(a)27.5v(b)86.5v(c)30v (d)42.5VAnswer:D

TOUGHQUESTIONS

1.ReversesaturationcurrentinsiliconPNjunctiondiodenearlydoublesforevery(a)20°Criseintemperature(b)50°Criseintemperature(c)60°Criseintemperature (d)10°CriseintemperatureAnswer:D

2.If,bymistakeacsourceinabridgerectifierisconnectedacrossthedcterminalsitwillburnoutandhence short___________diodes(a)One(b)Two(c)Three (d)FourAnswer:D

3.AVoltageof
200Vproducesadeflectionof90°inPMMCspringcontrolledinstrument.Ifthesameinstrumentsprovidedwithgravitycontrol,whatwouldbethedeflection?a)90° b)45° c)64.2° d)98°Answer:B

4.Inplateearthingtheearthplatemadeupofcoppersize

a)60cm*60cm*3.18mm

b)70cm*80cm*3.18mm

c)80cm*65cm*3.18mmd)90cm*60cm*3.18mmAnswer:A

5.Amovingcoilinstrumentgivesafullscaledetectionof20mA.Whenapotentialdifferenceof50mVis applied.Calculatetheseriesresistancetomeasure500Vonscale?a)2000ohm b)3000ohm c)3500ohm d)24997.5ohmAnswer:D

6.Theappliedinputacpowertoahalfwaverectifieris100watts.Thed.coutputpowerobtainedis40watts.Whatistherectificationefficiency?(a)10%(b)20%(c)30% (d)40%Answer:D

UNIT–4TRANSDUCERS

EASYQUESTIONS

1.Atransducerconvertinggroundmovementorvelocitytovoltageisknownas__________a)Geophoneb)Pickupc)Hydrophoned)SonartransponderAnswer:A

2.Whichistheexampleofaactivetransducer?a)Straingaugeb)Thermistorc)LVDTd)ThermocoupleAnswer:D

3. Whichtransducerisknownas‘self-
generatingtransducera)Activetransducerb)Passivetransducerc)Secondarytransducerd)AnalogtransducerAnswer:A

4. Whatistherelationbetweenscalefactorandsensitivityofatransducer?

a)Scalefactorisdoubleofsensitivity

b)Scalefactorisinverseofsensitivityc)Sensitivityisinverseofscalefactord)SensitivityisequaltoscalefactorAnswer:B

5.Whichofthefollowingisananalogtransducer?

a)Encodersb)Straingaugec)Digitaltachometersd)LimitswitchesAnswer:B

6.WhatistheprincipleofoperationofLVDT?a)Mutualinductanceb)Self-
inductancec)Permanenced)ReluctanceAnswer:A

7.WhichofthefollowingcanbemeasuredusingPiezo-electrictransducer?

a)Velocityb)Displacementc)Forced)SoundAnswer:C

8.Capacitivetransducerareusedfor?

a)Staticmeasurementb)Dynamicmeasurementc)Transientmeasurementd)BothstaticanddynamicAnswer:B

.9.Whichofthefollowingisusedinphotoconductivecell?
a)Seleniumb)Quartzc)Rochellesaltd)LithiumsulphateAnswer:A
10.Mechanicaltransducerssense__________a)electricalchanges

b)physicalchangesc)chemicalchangesd)biologicalchangesAnswer:B
11.Mechanicaltransducersgenerate________a)electricalsignalsb)chemicalsignalsc)physicalsignalsd)biologicalsignalsAnswer:C
12.Electricaltransducersgenerate________a)biologicalsignalsb)chemicalsignalsc)physicalsignals+d)electricalsignalsAnswer:D
13.Thepowerneedsofelectricaltransducersis________a)maximumb)minimumc)zerod)infiniteAnswer:B
14.Electricaltransducersare________a)smallandnon-portableb)largeandnon-portablec)smallandcompactd)largeandportableAnswer:C
15.PotentiometertransducersareusedforthemeasurementofA.PressureB.DisplacementC.HumidityD.Both(a)and(b)Answer:D
16.Thermistorisatransducer. ItstemperaturecoefficientisA.Negative

B.PositiveC.ZeroD.UniqueAnswer:A
17.StraingaugeisaA.ActivedeviceandconvertsmechanicaldisplacementintoachangeofresistanceB.PassivedeviceandconvertselectricaldisplacementintoachangeofresistanceC.PassivedeviceandconvertsmechanicaldisplacementintoachangeofresistanceD.ActivedeviceandconvertselectricaldisplacementintoachangeofresistanceAnswer:C
18.ThelinearvariabledifferentialtransformertransducerisA.InductivetransducerB.Non-inductivetransducerC.CapacitivetransducerD.ResistivetransducerAnswer:A
MODERATEQUESTIONS
1.Withtheincreaseintheintensityoflight,theresistanceofaphotovoltaiccellA.IncreasesB.DecreasesC.RemainssameD.DoubledAnswer:B
2.
IfthedisplacementismeasuredwithstraingaugethenthenumberofstraingaugenormallyrequiredareA.OneB.Two C.ThreeD.FourAnswer:D
3.LEDsfabricatedfromthegalliumarsenidesmitradiationintheA.VisibleRangeB.InfraredRegionC.UltravioletRegionD.UltrasonicRegion

Answer:B
4.Inlightemittingdiode,theavailablelightemittingregionisA.Lessthan2.5mmB.From2.5to25mmC.Greaterthan25mmD.Greaterthan50mmAnswer:B
5.Inliquidcrystaldisplays,theliquidcrystalexhibitspropertiesofA.LiquidB.SolidsC.GasesD.Both(a)and(b)Answer:D
6.Theopticalpropertiesofliquidcrystalsdependonthedirectionofa)Airb)Solidc)Lightd)WaterAnswer:C
7.LCDsoperatefromavoltagerangesfroma)3to15Vb)10to15Vc)10Vd)5VAnswer:A
8.LCDsoperatefromafrequencyrangesfroma)10Hzto60Hzb)50Hzto70Hzc)30Hzto60Hzd)NoneoftheMentionedAnswer:C

9.WhatisbackplaneinLCD?
a)Theacvoltageappliedbetweensegmentandacommonelementb)Thedcvoltageappliedbetweensegmentandacommonelementc)TheamountofpowerconsumedAnswer:A

10.
Inphotoemissivetransducers,electronsareattractedby___________a)Cathodeb)Anodec)Gridd)BodyAnswer:B
11.LDR'sarealsocalled_______________a)Photovoltaiccellb)Photoresistivecellc)Photoemissivecelld)AllofthementionedAnswer:B
12.Indark,LDRhasA. lowresistanceB.highcurrentC.highresistanceD.bothAandBAnswer:C
TOUGHQUESTIONS
31.1eVisequaltoA.1.6×10^{-19} JB.2.0×10^{-20} JC.3JD.4JAnswer:A
32.Solarcellworksbasedon(a)Lasertechnology(b)Photo-conduction(c)Thermalemission(c)TyndalleffectAnswer:B
33.Commonlyusedphotoemissivematerialis_______a)goldb)opiumc)telluriumd)cesium-antimonyAnswer:D

34.Photoconductorsaremadeof_______a)thicklayerofsemiconductorb)thinlayerofsemiconductorc)capacitivesubstrated)inductivesubstrateAnswer:B
35.Adevice
consistsofaphototransistorandaledisA.PhotodiodeB.OptocouplerC.OptoisolatorD.PhotomultiplierAnswer:B36.Aloadcellisessentiallya(a)straingauge (b)thermister
(c)resistivepotentiometer(d)inductivetransducerAnswer:A

UNIT–5DIGITALSYSTEMS
EASYQUESTIONS1.Communicationisthetransferofmeaningfulinformationfrom(a)sourcetodestination(b)transmittertoreceiver(c)sendertoreceiver (d)aboveallANSWER:D
2.Thebasicprocessofinformationexchangebetweentransmitterandreceiverisknownas…..
(a)communication (b)controlling(c)signaling(d)modulatingANSWER:A
3.Theprocessofconvertingelectricalequivalentoftheinformationtoasuitableformisdoneby….
(a)transmitter (b)receiver(c)medium(d)aboveallANSWER:A
4.Thecommunicationsystemwithwireasconductingmediumisknownas……
(a)wiredcommunication(b)linecommunication(c)guidedmediacommunication (d)aboveallANSWER:D
5.Thecommunicationsystemwhichhasnowiresasconductingmediumisknownas…..
(a)wirelesscommunication(b)radiocommunication(c)unguidedcommunication (d)aboveallANSWER:D
6.Noiseisbasicallya……(a)randomsignal(b)unwantedelectricalsignal(c)disturbancesignal
(d)aboveallANSWER:D
7.Theprocessofvaryingamplitudeofsinewavecarriersignalaccordingtotheinstantaneousvoltageofsin

ewavemodulatingsignalisknownas….
(a)FrequencyModulation(b)Phasemodulation(c)Amplitudemodulation (d)PAMANSWER:C

8.ThelossofinformationinAMwaveisknownas…(a)undermodulation
(b)overmodulation(c)attenuation(d)rectificationANSWER:B
9.Eachproducttermofagroup,a’.b.c’anda.b,representsthe____________inthatgroup.a)Inputb)POSc)Sum-of-Mintermsd)SumofMaxtermsANSWER:C
10.Each“1”entryinaK-mapsquarerepresents:a)AHIGHforeachinputtruthtableconditionthatproducesaHIGHoutputb)AHIGHoutputonthetruthtableforallLOWinputcombinationsc)ALOWoutputforallpossibleHIGHinputconditionsd)ADON’TCAREconditionforallpossibleinputtruthtablecombinationsANSWER:A
11.Whichofthefollowingexpressionsisinthesum-of-productsform?a)(A+B)(C+D)b)(A*B)(C*D)c)A*B*(CD)d)A*B+C*DANSWER:D
12.K-Mapoffulladderisof------variablesA.2b.3c.4d.1ANSWER:B
13.Theoutputofalogicgateis1whenallitsinputsareatlogic1,thegateiseither(a)ANANDoraNOR(b)AnANDoranOR(c)AnORoranX-OR(d)AnANDoraNORANSWER:B
14.Theoutputofalogicgateis1whenallitsinputsareatlogic0.Thegateiseither(a)ANANDoraNOR(b)AnANDoranX-OR(c)AnORoraNAND(d)AnX-ORoranX-NOR

ANSWER:A
15.Themostsuitablegatetocheckwhetherthenumberof1’sinadigitalwordisevenoroddis(a)X-OR
(b)NAND(c)NOR(d)AND,ORandNOTANSWER:A
16.Thenumberofrowsinthetruthtableofa4-inputgateis,(a)4(b)8(c)12 (d)16ANSWER:D
17.Forcheckingtheparityofadigitalword,itispreferabletouse(a)ANDgates(b)NANDgates (c)X-ORgates (d)NORgatesANSWER:C
18.A+AB+ABC+ABCD+ABCDE….=(a)1 (b)A (c)A+AB(d)ABANSWER:B
MODERATEQUESTIONS
1.AswithchingfunctionF(a,b,c,d)=a’b’cd+a’bc’d+a’bcd+ab’c’d+ab’cda.∑m(1,2,4,5,7)b.∑m(3,5,7,9,13)c.∑m(3,5,7,9,11)d.∑m(3,7,9,11,13)ANSWER:C
2.ThefunctionF(a,b,c,d)=∑m(5,9,11,14)isequivalenttoa.a’bc’d+ab’c’d+ab’cd+abcd’b.a’b’c’d+ab’c’d’+ab’cd+ab’cd’c.a’bc’d+ab’c’d+abcd+ab’cd’d.a’bc’d+a’b’c’d+ab’cd+a’bcd’ANSWER:A
3.IfSOPformofthefunctionF=a’bc’d+ab’c’d+abcd+ab’cd’a.F=(a+b’+c+d’)(a’+b+c+d’)(a’+b’+c’+d’)(a’+b+c’+d)b.F=(a+b’+c+d)(a’+b’+c+d’)(a’+b’+c’+d’)(a’+b’+c’+d)c.F=(a’+b’+c+d’)(a’+b+c+d’)(a+b’+c’+d’)(a’+b+c+d)d.F=(a+b’+c’+d’)(a’+b+c+d’)(a’+b’+c’+d’)(a’+b+c’+d’)ANSWER:A
4.Ifa3variablefunctionisrepresentedinPOSformasπ
M(0,3,6,7)theninSOPfromitisrepresentedasa.∑m(1,2,4,6)b.∑m(1,3,4,5)c.∑m(1,2,4,5)d.∑m(1,2,4,7)
ANSWER:C

5.Q.96.A+B=B+A;AB=BArepresentwhichlaws(a)Commutative(b)Associative(c)Distributive(d)IdempotenceANSWER:A
6. TheK-mapbasedBooleanreductionisbasedonthefollowingUnifyingTheorem:A+A’=1.a)Impactb)NonImpactc)For

ced)ComplementarityANSWER:B
7.Theprimeimplicantwhichhasatleastoneelementthatisnotpresentinanyotherimplicantisknownas____ _______a)EssentialPrimeImplicantb)Implicantc)Complementd)PrimeComplementANSWER:A
8.Product-of-Sumsexpressionscanbeimplementedusing___________a)2-levelOR-ANDlogiccircuitsb)2-levelNORlogiccircuitsc)2-levelXORlogiccircuitsd)Both2-levelOR-ANDandNORlogiccircuitsANSWER:D
9.Therearemanysituationsinlogicdesigninwhichsimplificationoflogicexpressionispossibleintermsof XORand_________________operations.a)X-NORb)XORc)NORd)NANDANSWER:A
10.Theselogicgatesarewidelyusedin_______________designandthereforeareavailableinICform.a)Samplingb)Digitalc)Analogd)Systems

ANSWER:B
11.Incellulartransmittersystem,thecarriergeneratedbyfrequencysynthesizerusesfollowingmodulationbytheamplifiedvoicesignalfrommicrophone(a)Frequencymodulation
(b)Phasemodulation(c)AMmodulation(d)NoneofaboveANSWER:B
12.Themodulationindexcorrespondingtomaximumdeviationandmaximummodulatingfrequencyiscalledas…
(a)modulationindex (b)deviationratio(c)pre-emphasisfactor(d)de-emphasisfactorANSWER:B
TOUGHQUESTIONS
1)Reducetheexpressiony=a’b’c’d+a’bc’d+a’bcd+a’bcd’+abc’d’+abc’d+abcd+ab’cda)acd+a’cd+ab’c+a’b’c’b)a’c’d+a’bc+abc’+acdc)a’c’d+abc+abc’+a’c’d’d)ac+a’bc+abc’+acdANSWER:B
2)simplifythefunctionf(a,b,c)=∑m(0,3,4,7)a)b’c’+bcb)a’b’+bcc)a’b’+abd)ab’+bcANSWER:A
3) (A+B)(A’*B’)=?a)1b)0c)ABd)AB’ANSWER:B
4.SimplifyY=AB’+(A’+B)C.a)AB’+Cb)AB+ACc)A’B+AC’d)AB+A

ANSWER:A
5.ThebooleanfunctionA+BCisareducedformof____________a)AB+BCb)(A+B)(A+C)c)A’B+AB’Cd)(A+C)BANSWER:B
6.Thecanonicalsumofproductformofthefunctiony(A,B)=A+Bis__________a)AB+BB+A’Ab)AB+AB’+A’Bc)BA+BA’+A’B’d)AB’+A’B+A’B’ANSWER:B